IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

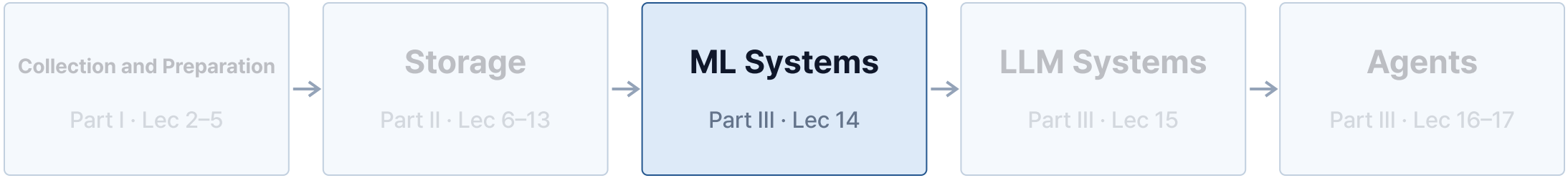
14.

ML Systems

Sangwoo Mo · POSTECH

Industrial & Management Engineering

Where We Are



Part III connects stored data to predictions, answers, and actions

Outline: ML Systems

01

Preparing Examples

use what was known at prediction time

02

Making Predictions

keep the input meaning the same

03

Checking Results

compare predictions with observed outcomes

01 · Preparing Examples

01

Preparing Examples

use what was known at prediction time

02

Making Predictions

keep the input meaning the same

03

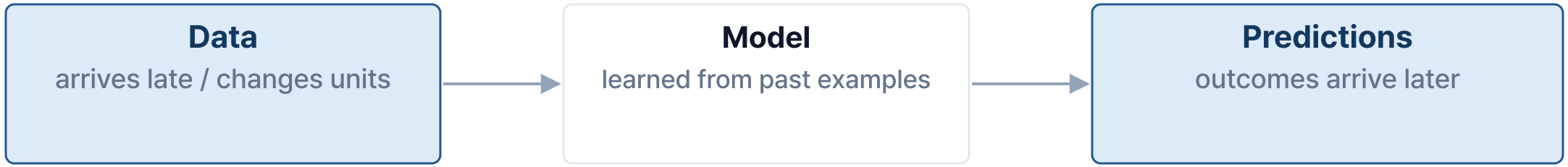
Checking Results

compare predictions with observed outcomes

Why ML Systems

A useful model also needs reliable data, predictions and feedback

2015 · Hidden Technical Debt in Machine Learning Systems

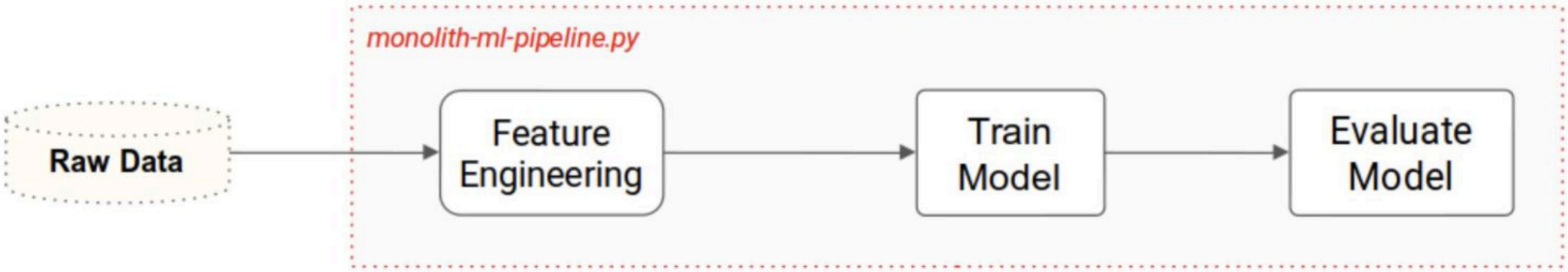


A good experiment still needs a dependable path to everyday predictions

The model is one component of a system that must keep working

From Data to a Checked Model

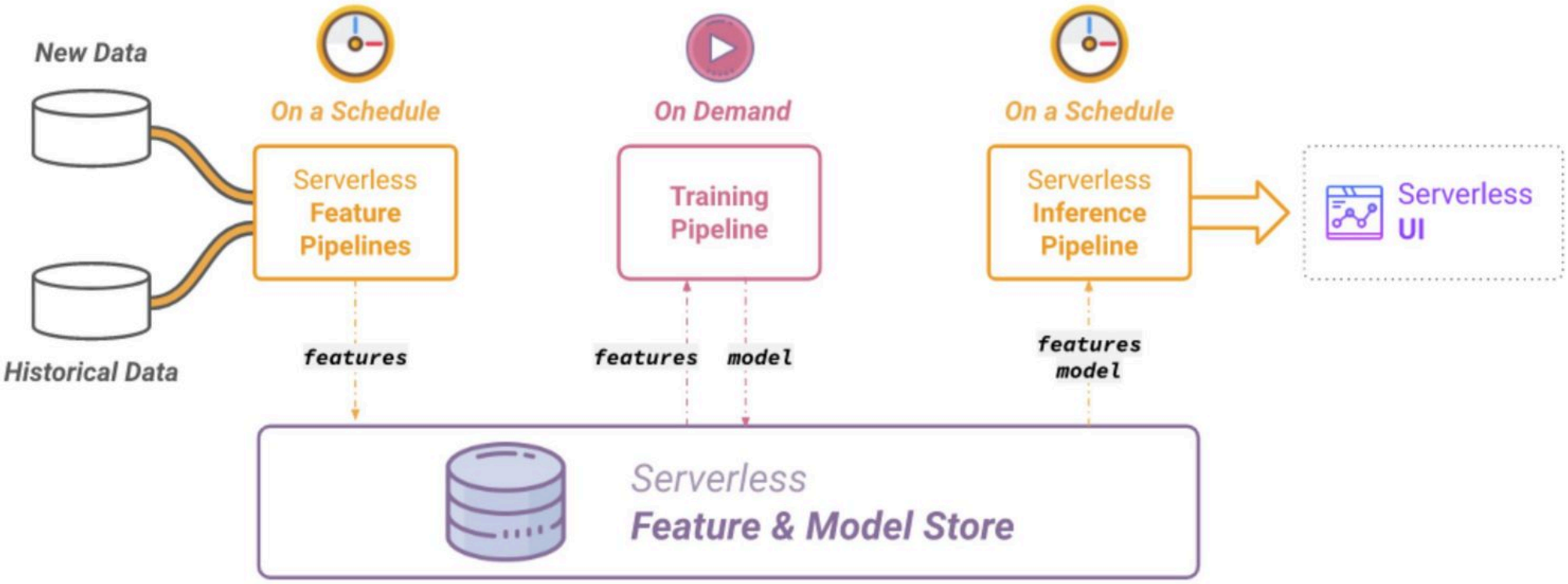
Prepare model inputs, learn from past outcomes, then check predictions



The pipeline connects data preparation to training and evaluation

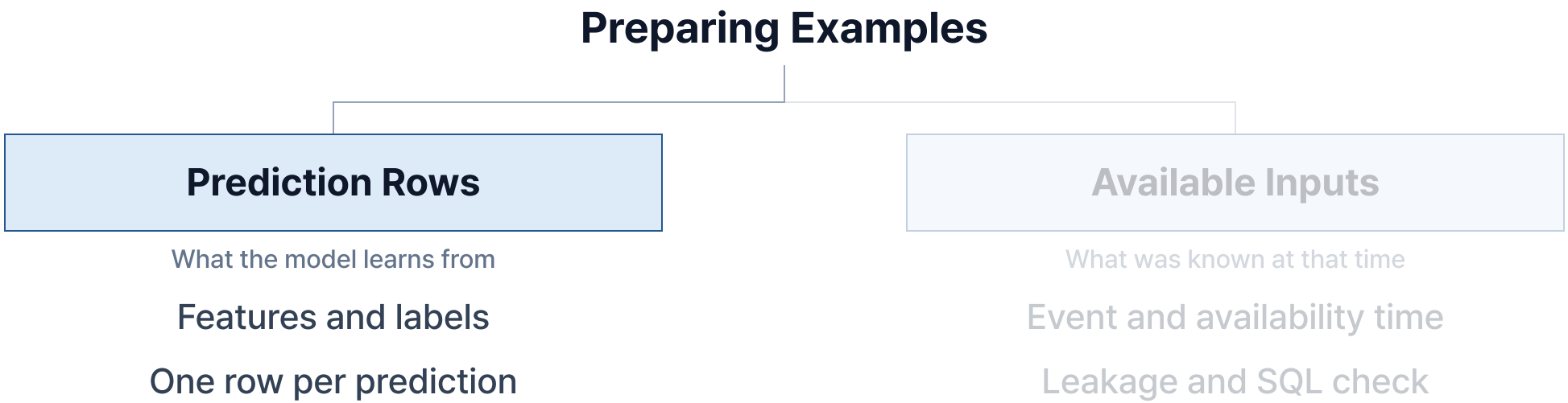
Separate ML Pipelines

Feature preparation, training and prediction can run on different schedules



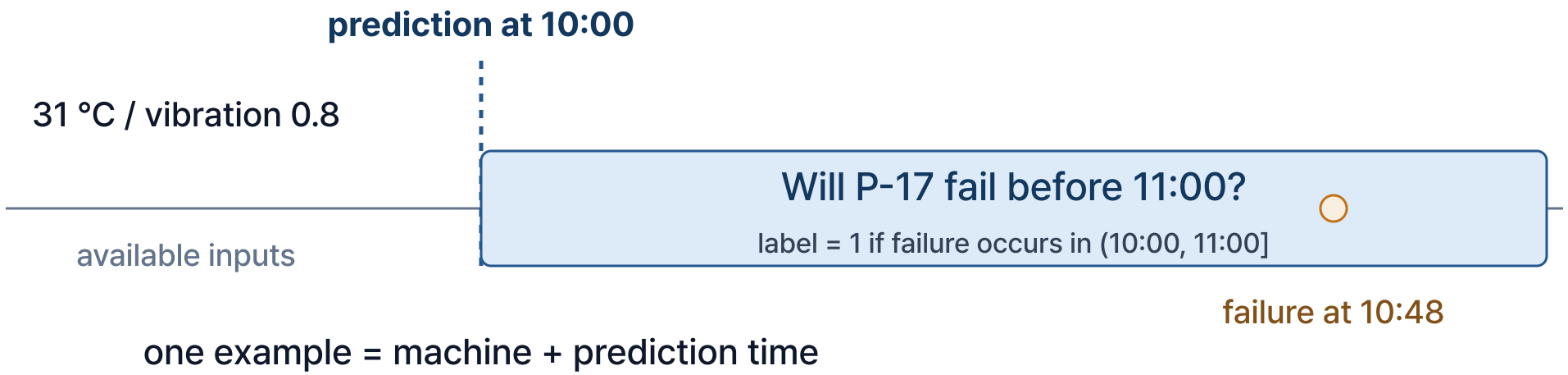
A shared feature definition connects pipelines that run at different times

§ Prediction Rows



Failure Prediction Example

At 10:00, predict whether P-17 will fail during the next hour

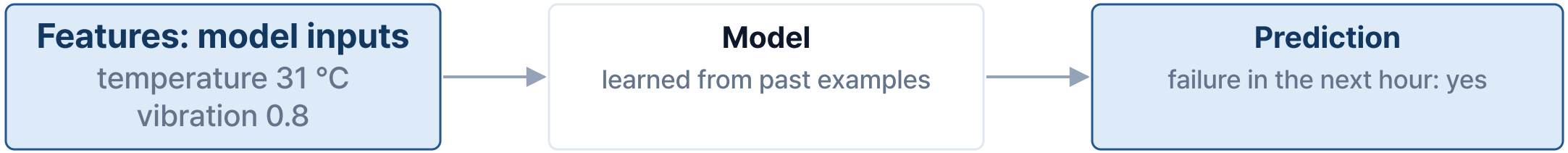


one example = machine + prediction time

The 10:48 failure becomes the label for the 10:00 prediction

Features and Labels

Features are model inputs and the label is the observed outcome

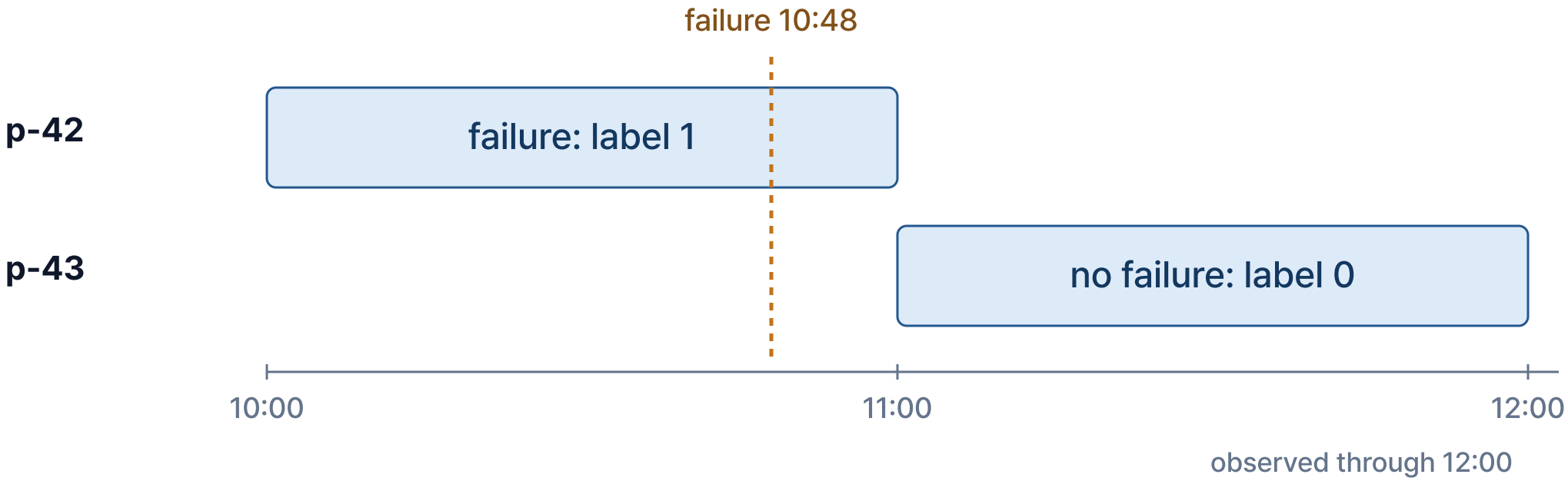


Label: the observed outcome arrives later

The model predicts a label that has not arrived yet

The Next-Hour Label

A label answers a question about one future time interval



The 10:48 failure belongs to the earlier prediction window

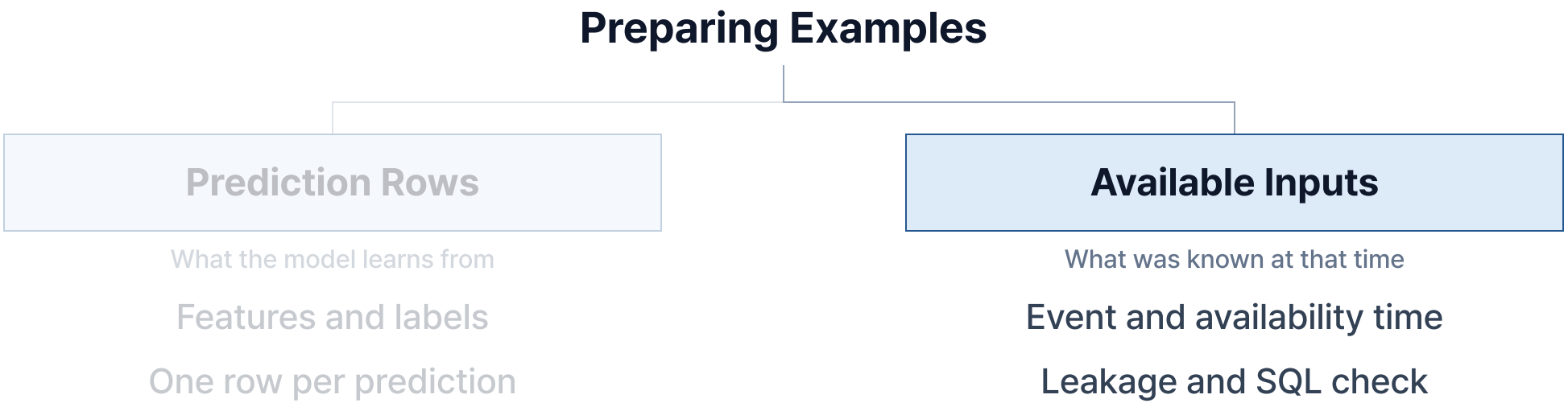
Training Examples

One row combines past inputs with the outcome that followed

Machine	Prediction time	Temperature	Vibration	Label
P-17	Sep 1 10:00	31 °C	0.8	1
P-17	Sep 1 11:00	28 °C	0.2	0
P-17	Sep 2 10:00	34 °C	0.9	1

Each row describes one P-17 prediction, rather than the whole machine

§ Available Inputs



Feature Availability

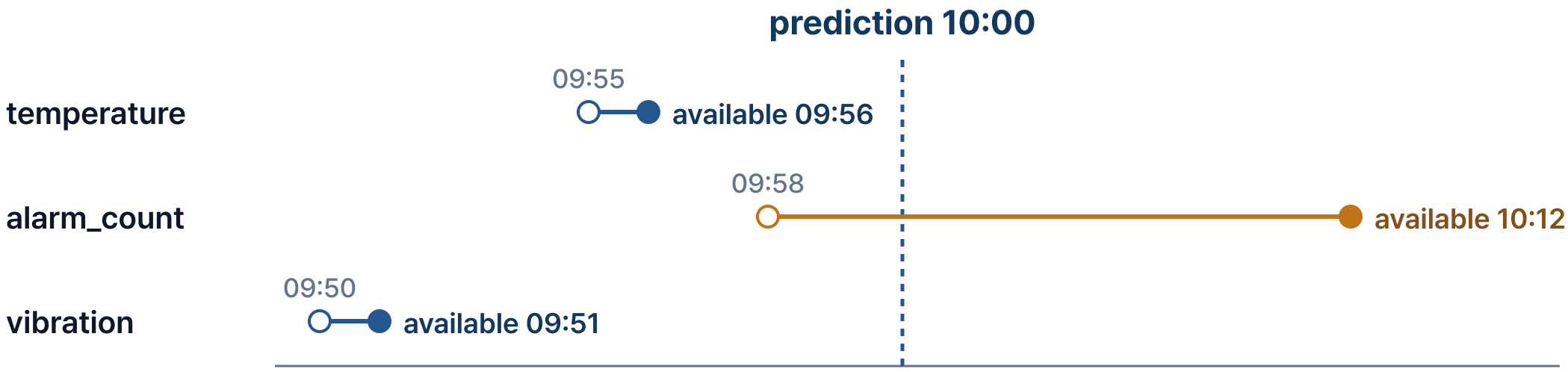
Event: when it happened. Available: when the system could read it

Feature	Event	Available	Value
temperature	09:55	09:56	31
alarm_count	09:58	10:12	3
vibration	09:50	09:51	0.8

The alarm happened before 10:00 but arrived at 10:12

Late Feature Arrival

An earlier event can still be unknown when the model predicts



Only temperature and vibration are available before the prediction line

Caveat: Future Information

Using a late input during training gives the model information it would lack

Training input at 10:00	Values	Available then?
temperature / vibration	31 / 0.8	yes
alarm_count	3	no, arrived 10:12

The late alarm would make the training example unrealistically informative

Demo: Available Inputs

Run the join, then exclude inputs that arrived after 10:00

1. Join Predictions and Features using machine_id; filter event_at ≤ predicted_at.
2. Also filter available_at ≤ predicted_at. Which feature disappears?

```
-- write your query here, then Run
```

QUERY RESULT

Write a query on the left, then Run

Answer: Available Inputs

Both time conditions must hold for the input to be usable

```
SELECT p.id, f.feature, f.value
FROM Predictions p JOIN Features f ON
f.machine_id=p.machine_id
WHERE f.event_at <= p.predicted_at AND
f.available_at <= p.predicted_at
ORDER BY p.id, f.feature;
```

id	feature	value
prediction_42	temperature	31.0
prediction_42	vibration	0.8

The late alarm report is excluded

Available Model Inputs

The filtered result reconstructs what the predictor could actually read

Prediction	Usable input	Value
P-17 at 10:00	temperature	31
P-17 at 10:00	vibration	0.8

The two remaining values are the inputs for the 10:00 prediction

02 · Making Predictions

01

Preparing Examples

use what was known at prediction time

02

Making Predictions

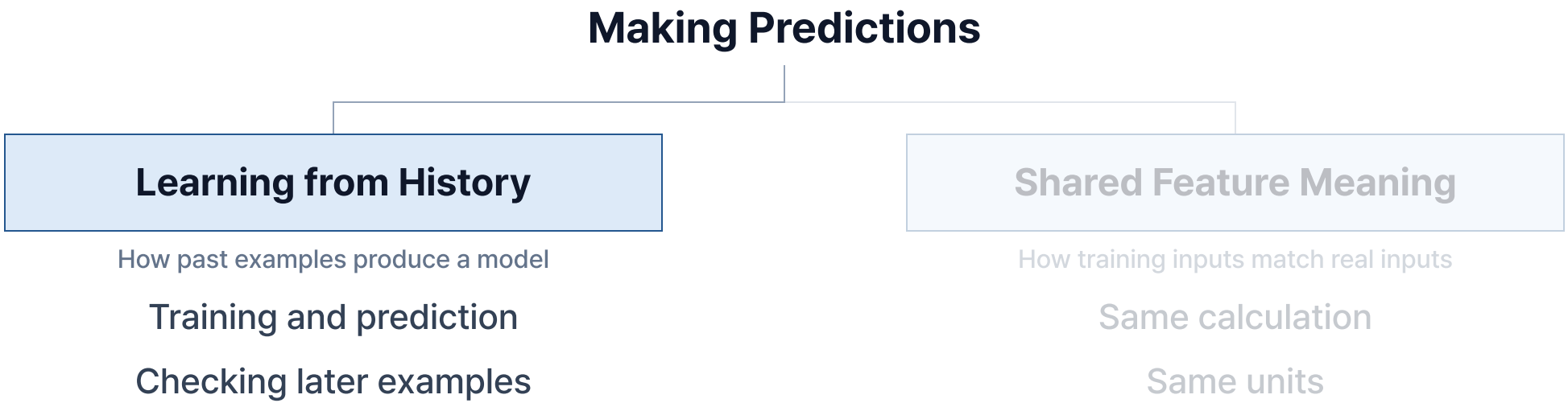
keep the input meaning the same

03

Checking Results

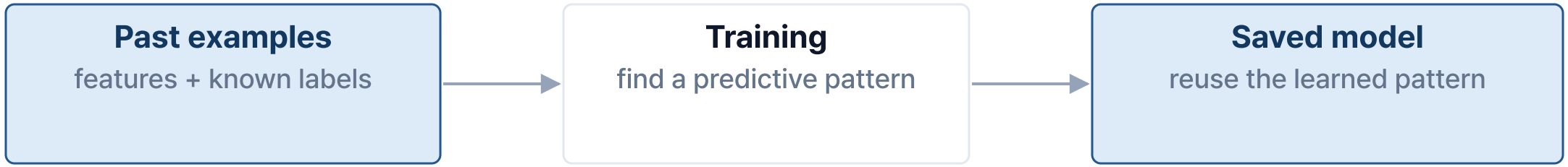
compare predictions with observed outcomes

§ Learning from History



Training and Prediction

Training learns from known outcomes and prediction uses the saved model

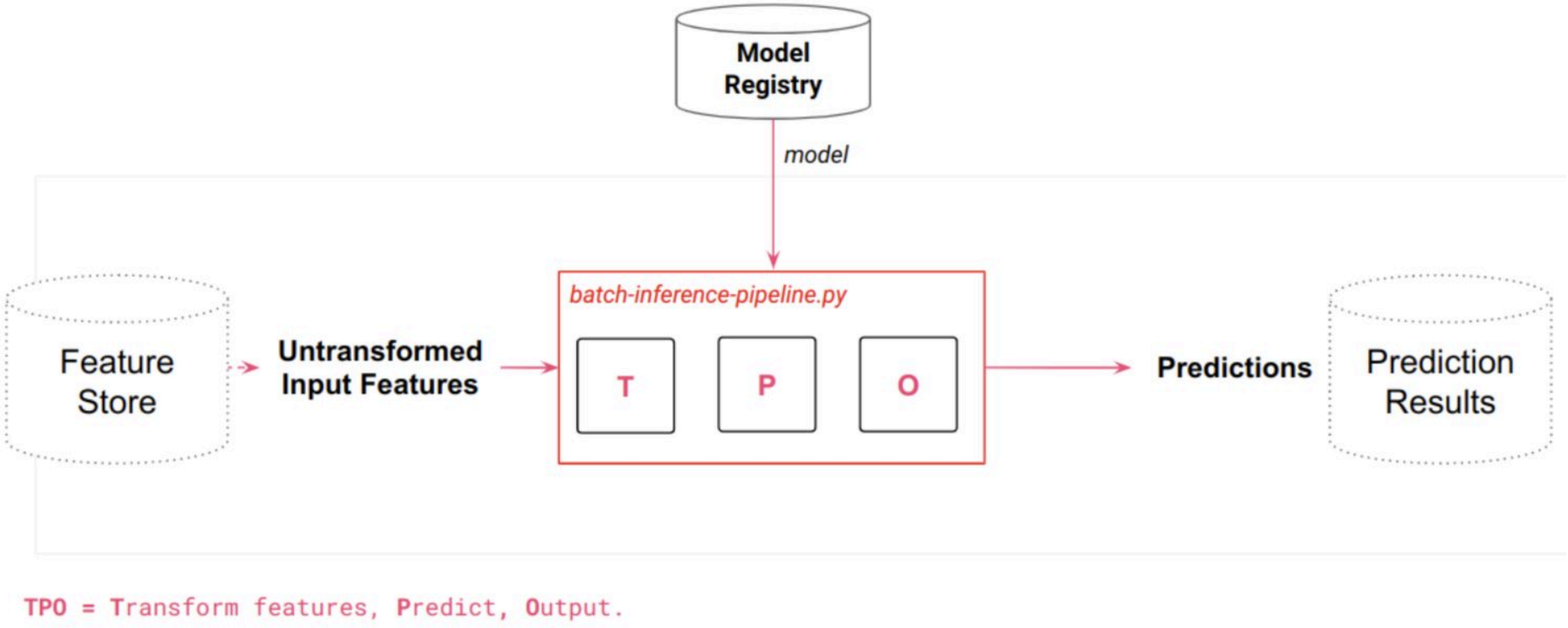


Prediction uses features while the future label is unknown

The model learns from past labels and predicts a future label

Batch Inference

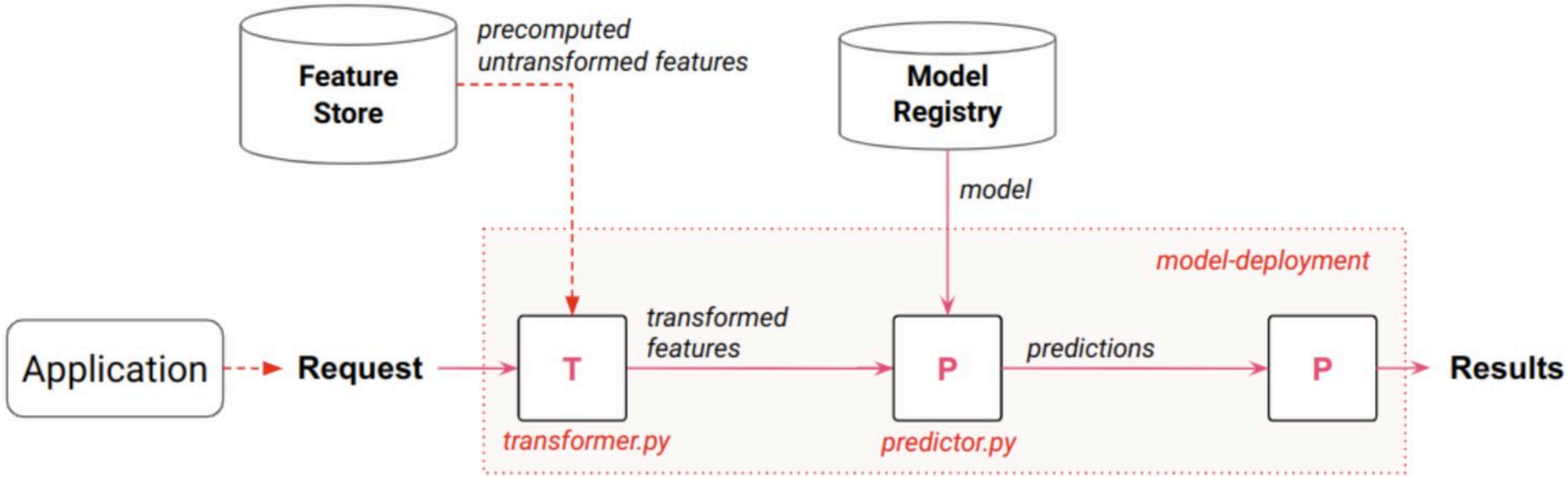
Compute a set of predictions ahead of the time they will be read



The pipeline writes a prediction table that applications can read later

Request-Time Inference

Calculate a prediction when a request arrives



TPP = Transform the input request into features, Predict using input features and the model, Post-process predictions, before output results.

Request-time serving needs both available features and the saved model

Training, Validation and Test

Fit on training data, select with validation, and reserve the test for the final check



The three sets have different jobs; the percentages are one illustrative split

Overlapping Label Windows

Splitting prediction rows can still let the same future failure cross the boundary

Train: 09:50
label until 10:50

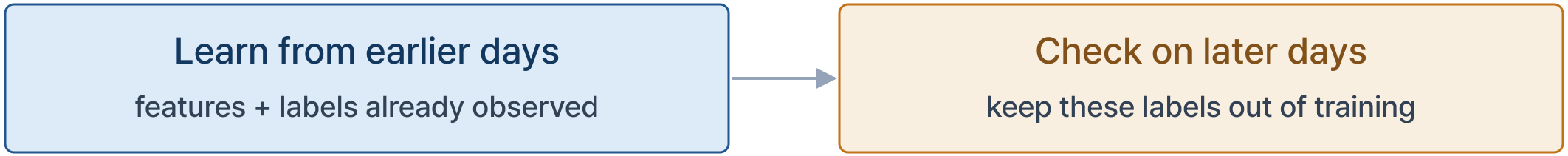
Split: 10:00
failure at 10:48

Test: 10:00
label until 11:00

The same 10:48 failure labels rows on both sides of the split

Checking Later Examples

Check the model on later examples that it did not learn from



P-17: train before Sep 8, check predictions from Sep 8 onward

Past training data and later checking data serve different purposes

Predictions and Outcomes

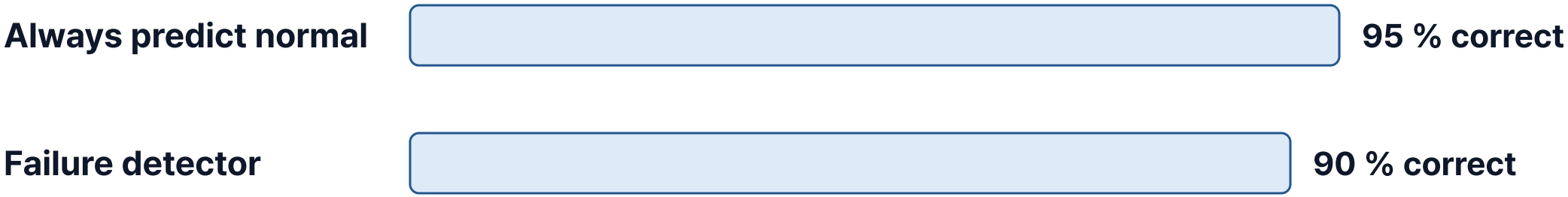
A prediction is correct only when it matches the observed label

P-17 prediction	Predicted failure	Observed label	Correct?
Sep 8 10:00	yes	1	yes
Sep 8 11:00	no	0	yes
Sep 9 10:00	no	1	no
Sep 9 11:00	no	0	yes

Three of the four predictions match their observed outcomes

Accuracy and Missed Failures

A high overall accuracy can hide the failures the system was built to detect




```
Always normal: 5 misses; detector: 2 misses + 8 false alarms
```

The 95% model misses every failure in this illustrative set of 100 cases

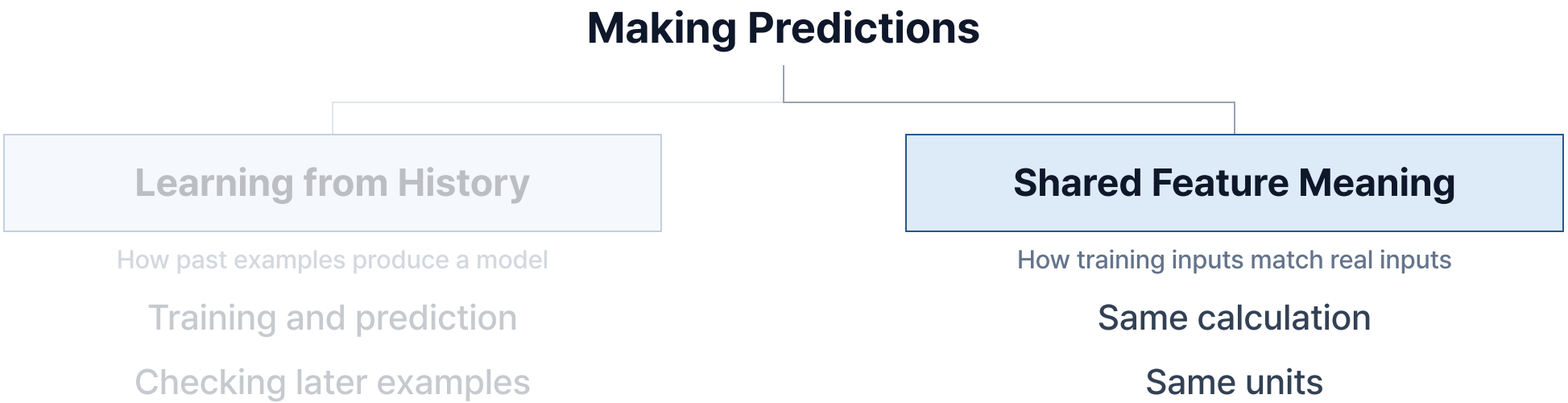
Error Costs and a Decision

Compare errors using the operational cost of each outcome

Prediction policy	Missed failures	False alarms	Illustrative total cost
Always normal	5	0	5 × 100 = 500
Failure detector	2	8	2 × 100 + 8 × 1 = 208

The preferred model changes when the costs of errors are included

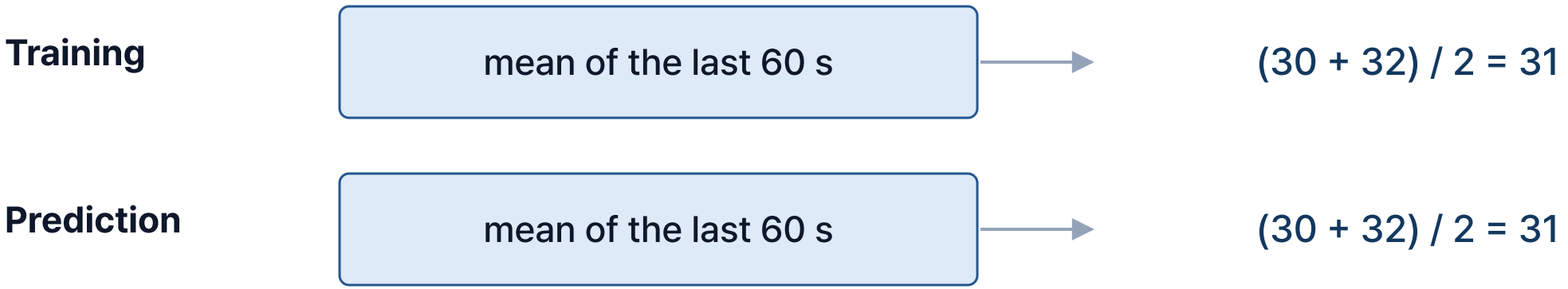
§ Shared Feature Meaning



A Temperature Feature

A feature definition tells us how source readings become one input

P-17 temperature readings: 30 °C at 09:59:10, 32 °C at 09:59:50



Both paths turn the same readings into the value 31

Caveat: Different Calculations

A last reading and a mean can share a name while carrying different meanings

P-17 temperature readings: 30 °C at 09:59:10, 32 °C at 09:59:50

Training	mean of the last 60 s	(30 + 32) / 2 = 31
Prediction	last reading	32

Training receives 31 while prediction receives 32 from the same readings

Shared Feature Meaning

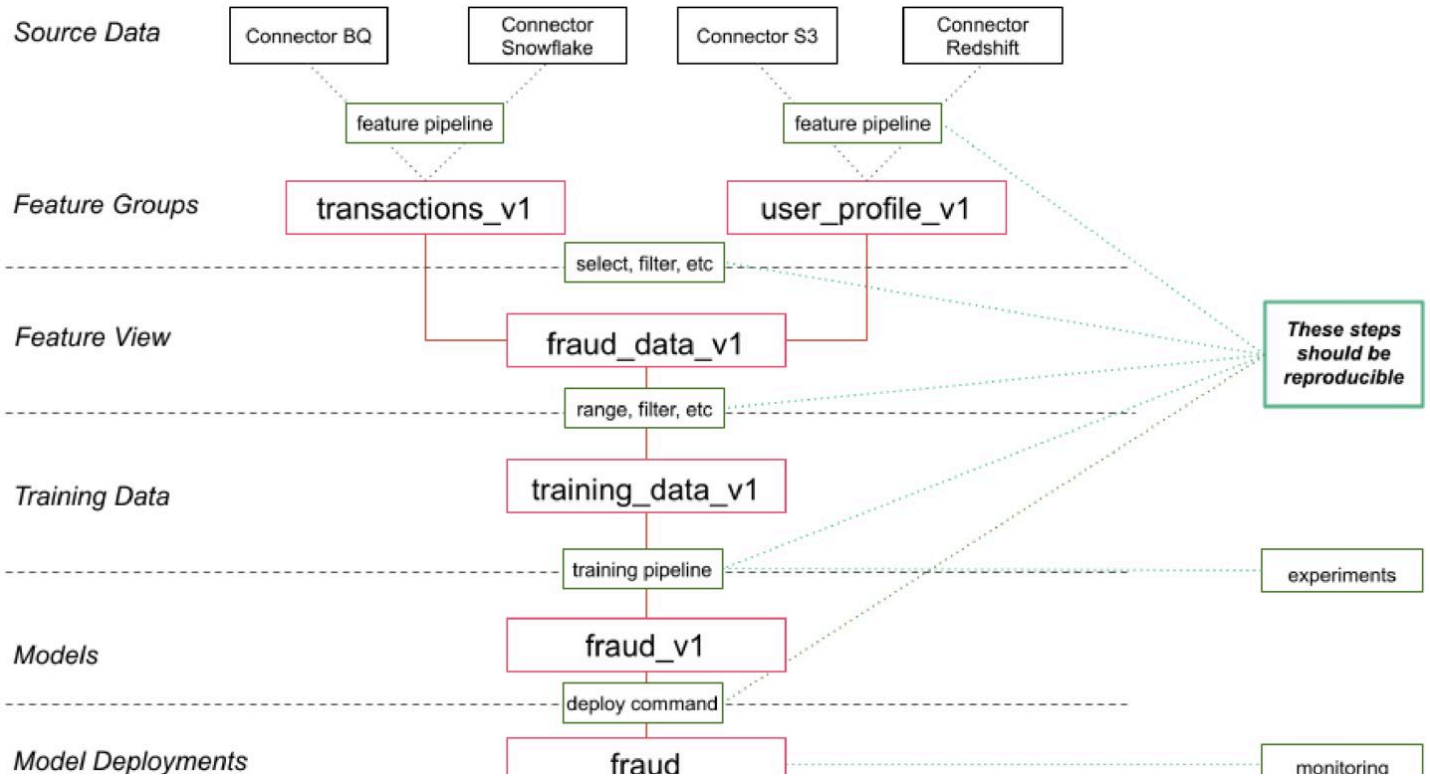
Keep the calculation and the unit the same in both paths

Path	Calculation	Unit	Input
Training	mean of last 60 s	°C	31
Prediction	mean of last 60 s	°C	31

The model receives the same kind of input during training and prediction

Reproducible ML Assets

Recover the data, feature definition and model version behind a prediction



A model file alone does not recover the inputs that produced it

The Same Weights, Different Inputs

Changing the feature rule can change the prediction without changing the model

Raw readings	Feature rule	Fixed decision rule
30°C, 32°C	mean 31 / last 32	alarm if x > 31.5

$$31 \le 31.5 \rightarrow \text{normal};\ 32 > 31.5 \rightarrow \text{alarm}$$

The mean produces normal; the last reading produces an alarm with unchanged weights

03 · Checking Results

01

Preparing Examples

use what was known at prediction time

02

Making Predictions

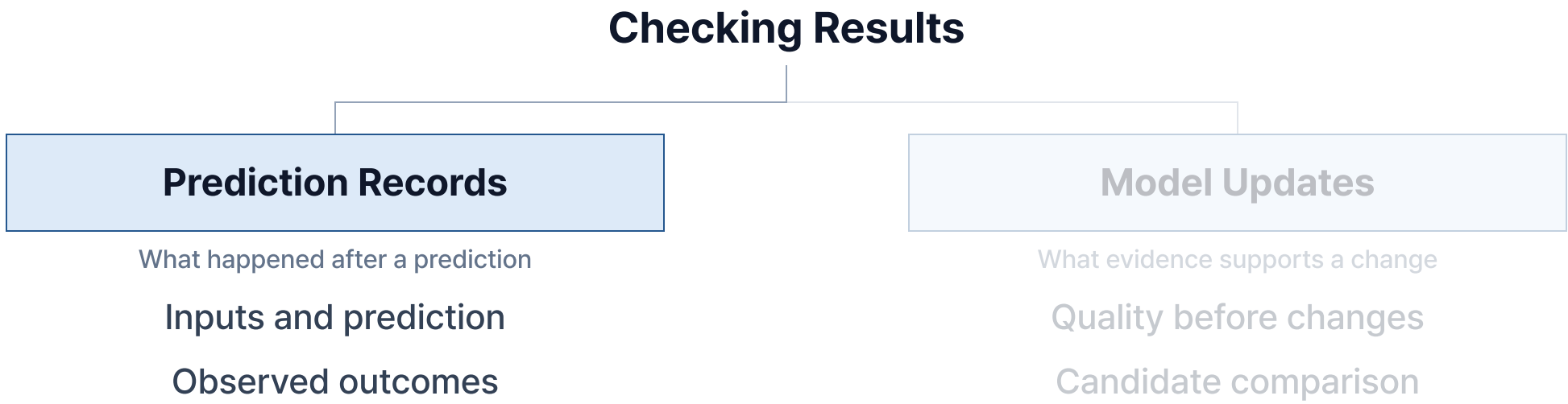
keep the input meaning the same

03

Checking Results

compare predictions with observed outcomes

§ Prediction Records



Prediction Record

Keep the input and prediction so the later outcome can be matched

Prediction ID	Machine / Time	Model inputs	Prediction	Observed label
p-42	P-17 / 10:00	31 °C / 0.8	failure: yes	?

The 10:00 prediction is recorded before its outcome is known

Observed Outcome

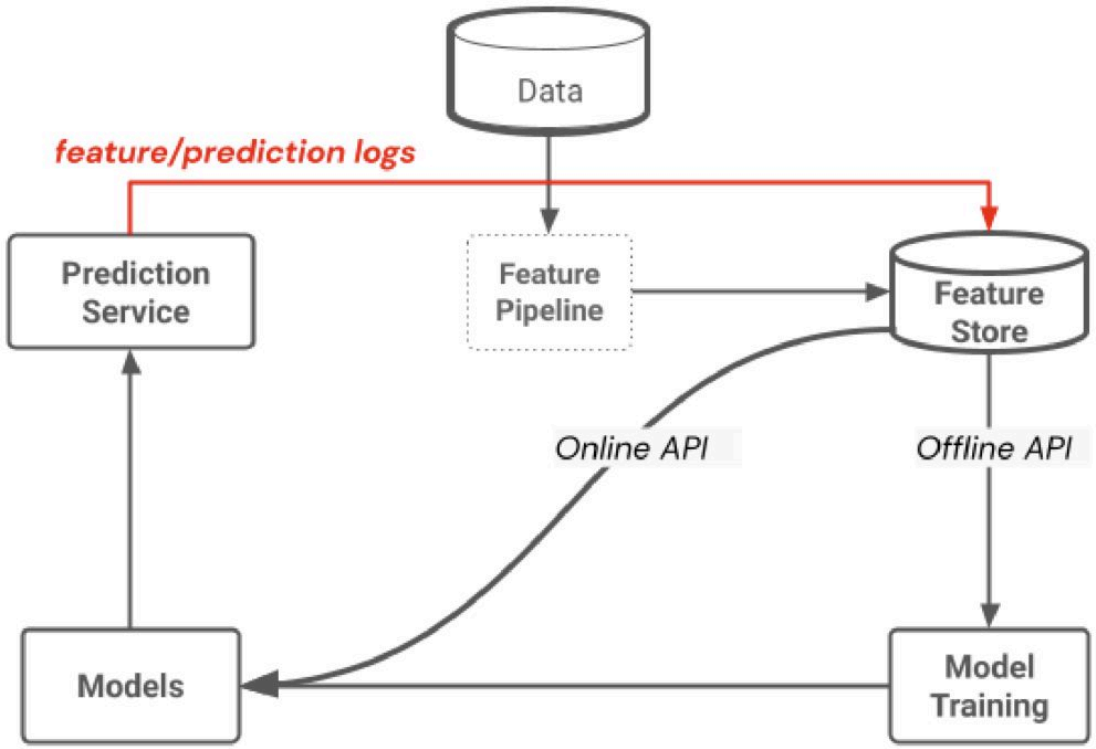
Attach the observed failure to the prediction it answers

Prediction ID	Machine / Time	Model inputs	Prediction	Observed label
p-42	P-17 / 10:00	31 °C / 0.8	failure: yes	1: failure

The 10:48 failure confirms the recorded prediction p-42

Feature and Prediction Logs

Keep evidence that can reconnect a prediction with its input and later label



Prediction logs let later outcomes be matched to the earlier decision

Delayed Outcomes

Recent predictions may still have missing outcomes

Ten P-17 predictions from the same day



Day 1: 4 labels, 4 correct

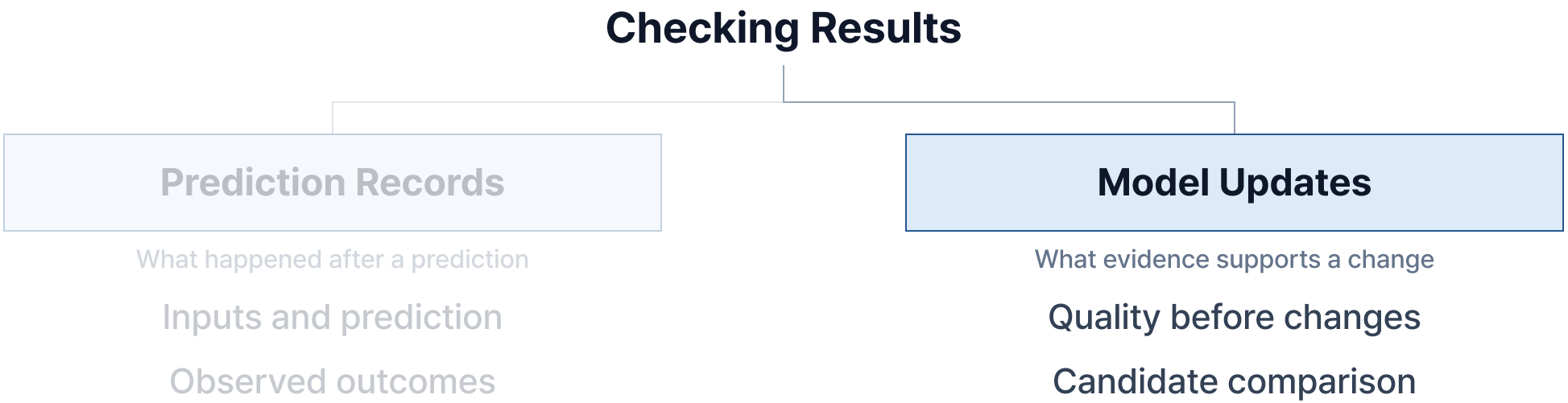
Only 4 outcomes are known

Day 3: 10 labels, 6 correct

All 10 outcomes are known

The same ten predictions score differently as more labels arrive

§ Model Updates



Input Changes and Quality

Check what changed before deciding to train a new model

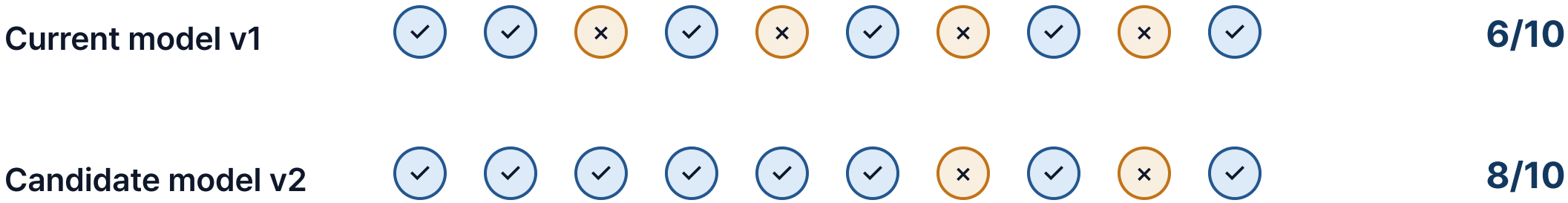
Check	Earlier	Now	Meaning
Temperature unit	°C	°F	fix input conversion
Correct predictions	6 / 10	6 / 10	quality unchanged

A unit change needs an input fix before any model update

Candidate Comparison

Compare the current model and a candidate on the same later examples

The same ten later predictions, with all labels observed



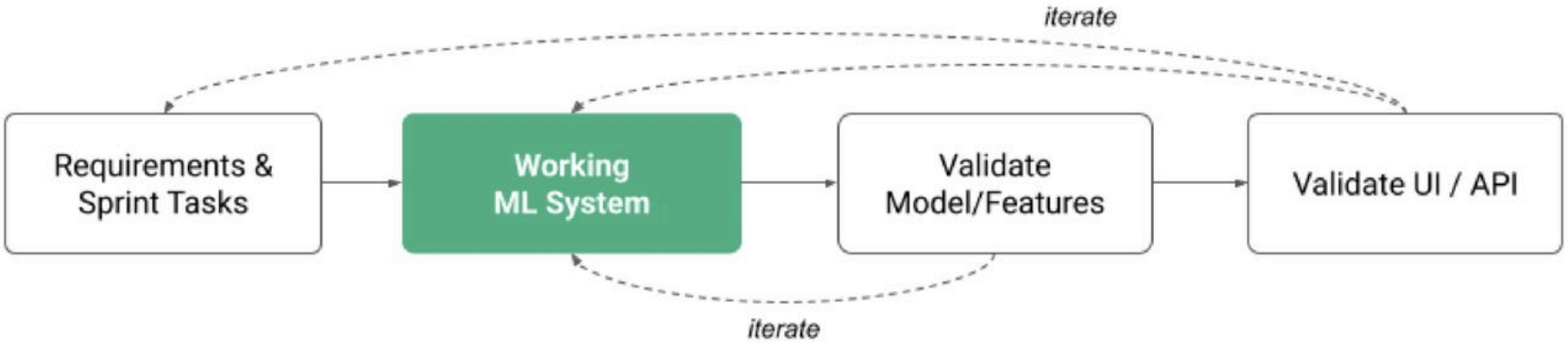
Each mark compares one prediction with its observed label

The candidate matches eight outcomes while the current model matches six

MLOps: A Repeating Cycle

Develop a model, use it, observe outcomes and improve the next candidate

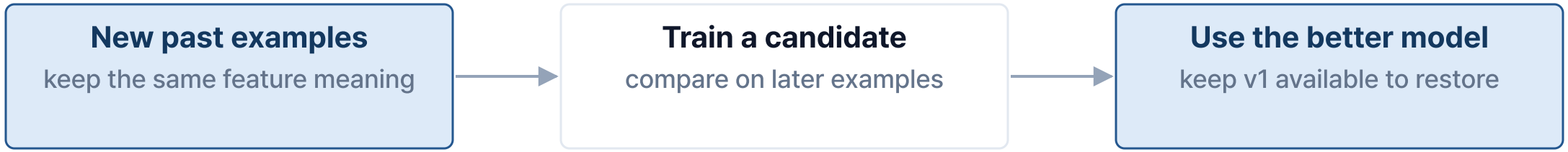
- A goal of MLOps is to improve both iteration speed and quality when developing ML



Operating the model continues after the first deployment

Model Update

Use observed examples to train and check a candidate before replacing the model



A checked candidate can replace v1 while v1 remains available

A Limited Model Release

Keep the baseline and compare observed outcomes before widening a release

Candidate
version v2 + feature v3

Limited release
same eligible machines

Rollback condition
error or cost rises

Model and feature versions move together through the release decision

Takeaway: Operate with available data

1. Reconstruct available inputs and keep train, validation and test roles separate
2. Preserve feature definitions and compare the costs of prediction errors
3. Link versions, predictions and mature labels before releasing an update